SET 21: ENERGY CONSUMPTION ANALYSIS

Dataset	Energy_Consumption_Data.csv
Department	AIDS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R
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Dataset: Energy_Consumption_Data.csv

Sector	Region	Month	Temperature (°C)	Units_Consumed (kWh)	Cost (₹)	Renewable_Usage (%)	Peak_Hours
Residential	North	Jan	15	320	2100	22	4
Commercial	South	Jan	24	540	3600	18	6
Industrial	West	Feb	20	880	5900	12	8
Residential	East	Feb	18	350	2300	25	5
Commercial	North	Mar	28	610	4100	20	7
Industrial	South	Mar	30	920	6200	15	9

PROGRAM 1: Histogram and Density Plot of Units Consumed

Question: Import the dataset in R. Plot a histogram and density plot for Units_Consumed. Compare consumption patterns.

```
# Import the dataset (the CSV file must be in the working directory)
energy <- read.csv("Energy_Consumption_Data.csv")
print(energy)

# Histogram (with density curve) and Density Plot side by side
par(mfrow = c(1, 2))

hist(energy$Units_Consumed,
     breaks = seq(300, 1000, by = 100),
     freq = FALSE,
     main = "Histogram of Units Consumed",
     xlab = "Units Consumed (kWh)",
     ylab = "Density",
     col = "skyblue")
lines(density(energy$Units_Consumed), col = "red", lwd = 2)

plot(density(energy$Units_Consumed),
     main = "Density Plot of Units Consumed",
     xlab = "Units Consumed (kWh)",
     ylab = "Density",
     col = "darkblue", lwd = 2)
polygon(density(energy$Units_Consumed), col = rgb(0, 0, 1, 0.3))

par(mfrow = c(1, 1))
```

```
# Compare consumption patterns by sector
aggregate(Units_Consumed ~ Sector, data = energy,
          FUN = mean)
```

```
  Sector = c("Residential", "Commercial", "Industrial",
             "Residential", "Commercial", "Industrial"),
  Region = c("North", "South", "West", "East", "North", "South"),
  Month = c("Jan", "Jan", "Feb", "Feb", "Mar", "Mar"),
  Temperature = c(15, 24, 20, 18, 28, 30),
  Units_Consumed = c(320, 540, 880, 350, 610, 920),
  Cost = c(2100, 3600, 5900, 2300, 4100, 6200),
  Renewable_Usage = c(22, 18, 12, 25, 20, 15),
  Peak_Hours = c(4, 6, 8, 5, 7, 9)
)

# Save it as a CSV in the working directory
write.csv(energy_data, "Energy_Consumption_Data.csv", row.names = FAL

# Import it
energy <- read.csv("Energy_Consumption_Data.csv")
```

```
18:50  (Top Level)                                    R Script
```

```
Console  Terminal  Background Jobs
R 4.5.0 · ~/
  Sector Region Month Temperature Units_Consumed Cost
1 Residential  North   Jan          15            320 2100
2  Commercial  South   Jan          24            540 3600
3  Industrial   West   Feb          20            880 5900
4 Residential   East   Feb          18            350 2300
5  Commercial  North   Mar          28            610 4100
6  Industrial  South   Mar          30            920 6200
  Renewable_Usage Peak_Hours
1              22          4
2              18          6
3              12          8
4              25          5
5              20          7
```

Output 1: Histogram and Density Plot of Units Consumed

Observation: Units consumed range from 320 to 920 kWh, with an average of about 603 kWh. The histogram and density curve are spread out and show two groups: a lower-consumption group (320–610 kWh) and a high-consumption group (880–920 kWh). By sector, Industrial uses the most (average 900 kWh), Commercial is in the middle (575 kWh) and Residential uses the least (335 kWh).

PROGRAM 2: Scatter Plot of Temperature vs Units Consumed

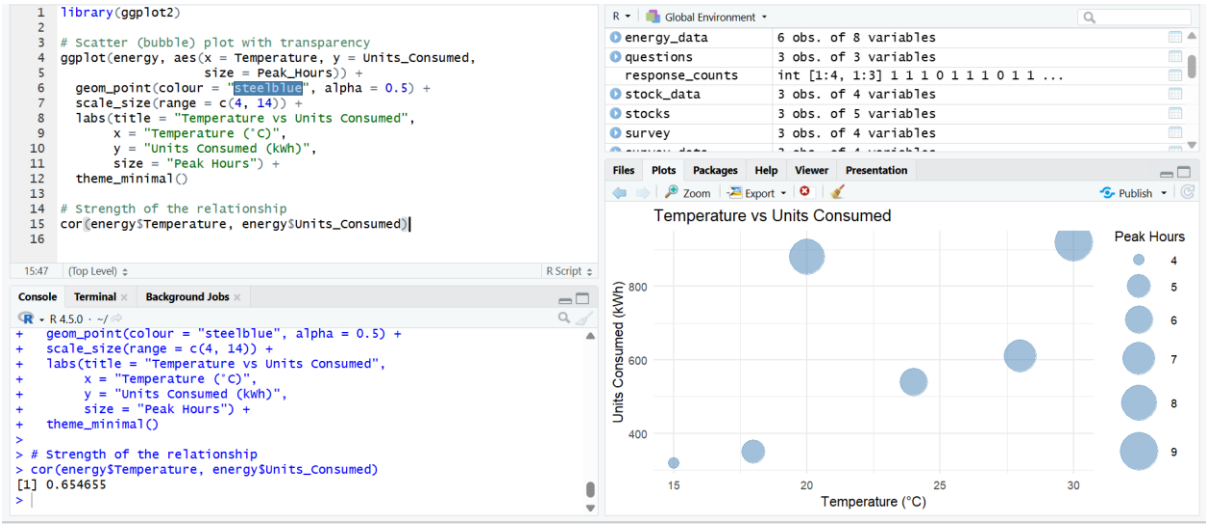
Question: Using R programming, create a scatter plot between Temperature and Units_Consumed. Represent Peak_Hours using bubble size. Apply transparency.

```
library(ggplot2)

# Scatter (bubble) plot with transparency
ggplot(energy, aes(x = Temperature, y = Units_Consumed,
                   size = Peak_Hours)) +
  geom_point(colour = "steelblue", alpha = 0.5) +
  scale_size(range = c(4, 14)) +
  labs(title = "Temperature vs Units Consumed",
```

```
        x = "Temperature (°C)",
        y = "Units Consumed (kWh)",
        size = "Peak Hours") +
  theme_minimal()

# Strength of the relationship
cor(energy$Temperature, energy$Units_Consumed)
```




```
> cor(energy$Temperature, energy$Units_Consumed)
[1] 0.6546550
```

Output 2: Scatter Plot with Bubble Size for Peak Hours

Observation: Units consumed generally increase as temperature rises (correlation of about 0.65, a moderate positive relationship). The larger bubbles, meaning more peak hours, appear at the higher consumption levels (880 and 920 kWh), so higher temperature and more peak hours go together with higher consumption. The transparency (alpha = 0.5) makes overlapping bubbles easy to see.

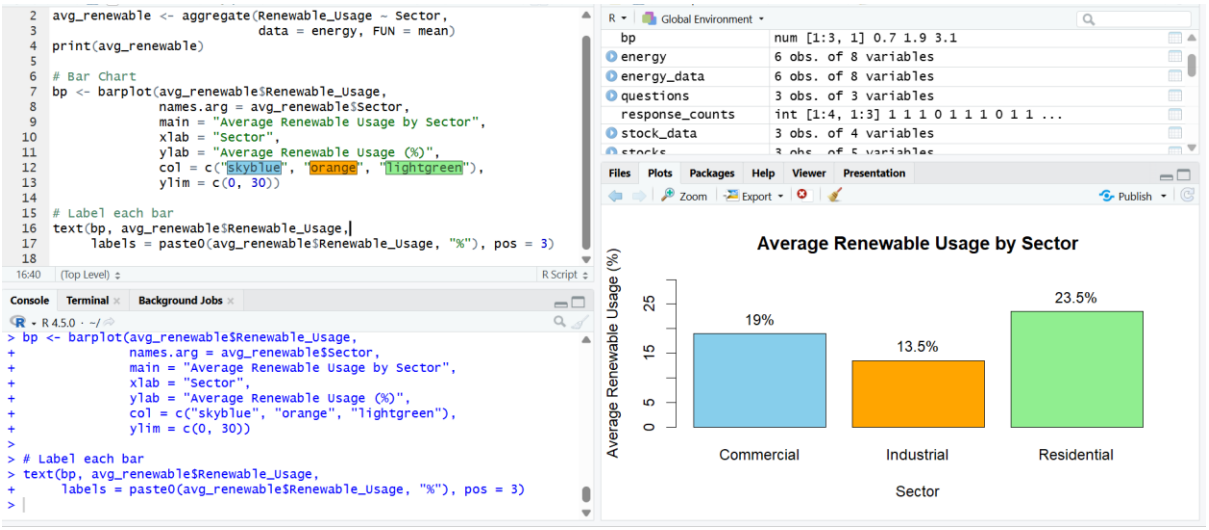
PROGRAM 3: Average Renewable Usage by Sector

Question: Using R, calculate average Renewable_Usage for each Sector. Create a bar chart. Interpret renewable energy adoption.

```
# Average Renewable_Usage for each Sector
avg_renewable <- aggregate(Renewable_Usage ~ Sector,
                           data = energy, FUN = mean)
print(avg_renewable)

# Bar Chart
bp <- barplot(avg_renewable$Renewable_Usage,
              names.arg = avg_renewable$Sector,
              main = "Average Renewable Usage by Sector",
              xlab = "Sector",
              ylab = "Average Renewable Usage (%)",
              col = c("skyblue", "orange", "lightgreen"),
              ylim = c(0, 30))

# Label each bar
text(bp, avg_renewable$Renewable_Usage,
     labels = paste0(avg_renewable$Renewable_Usage, "%"), pos = 3)
```



Output 3: Average Renewable Usage by Sector

Observation: Residential has the highest average renewable usage (23.5%), followed by Commercial (19%), while Industrial has the lowest (13.5%). Renewable energy adoption is strongest in the residential sector and weakest in the industrial sector, which is also the largest consumer of electricity, so industry has the greatest scope to increase its use of renewable energy.

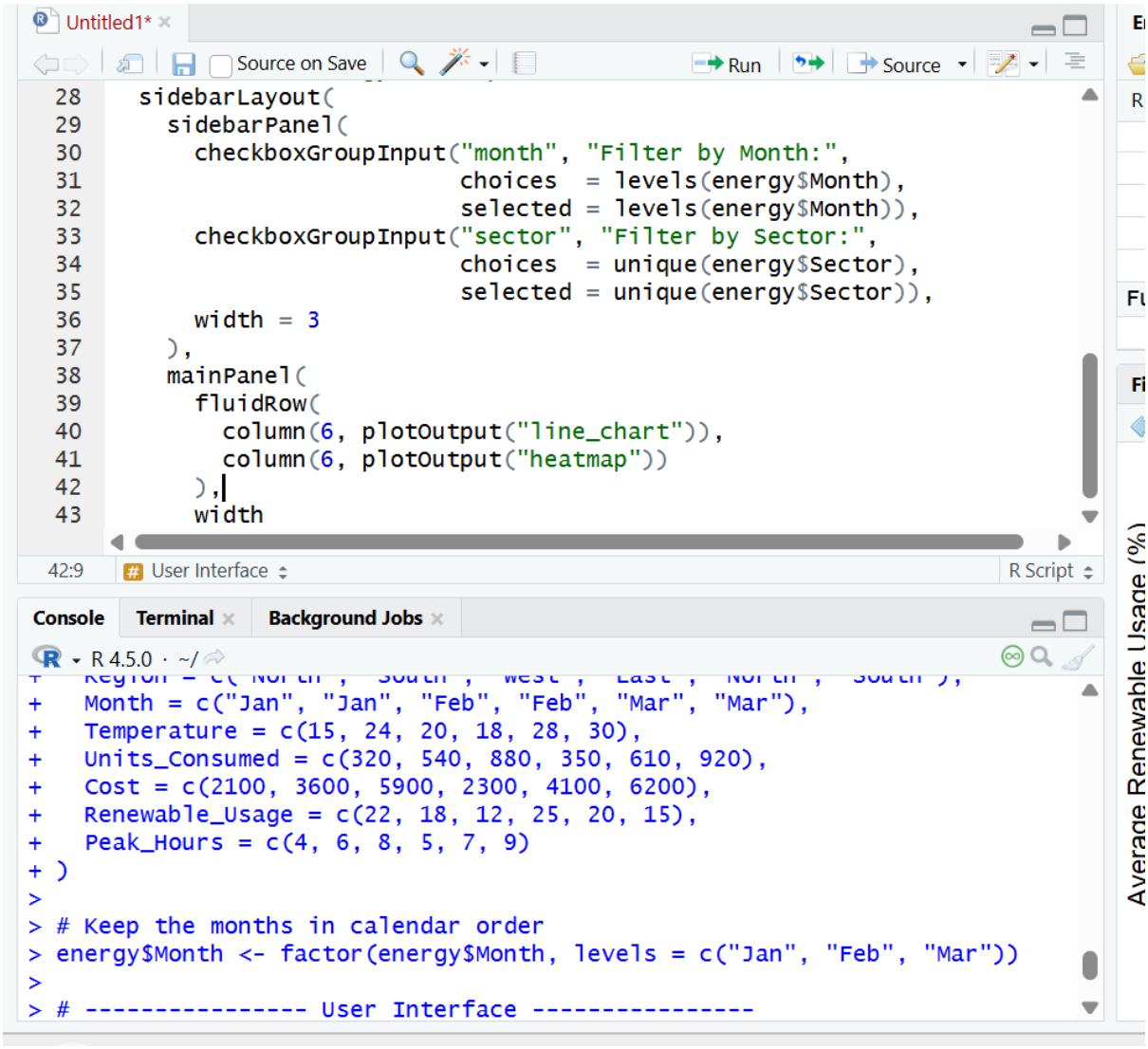
PROGRAM 4: Interactive Tableau Dashboard

Question: Import Energy_Consumption_Data.csv into Tableau. Create a line chart for monthly energy usage trends. Create a heatmap (Sector vs Region vs Units). Add filters for Month and Sector. Design an interactive dashboard.

Tableau Steps

1. Open Tableau and connect to the text file Energy_Consumption_Data.csv.
2. Verify the fields Sector, Region, Month, Temperature, Units_Consumed, Cost, Renewable_Usage and Peak_Hours.
3. Create a worksheet for the line chart: drag Month to Columns (sorted Jan, Feb, Mar) and SUM(Units_Consumed) to Rows; drag Sector to Color; select Line in the Marks card.
4. Add the title “Monthly Energy Usage Trend” and label the axes.
5. Create a second worksheet for the heatmap: drag Region to Columns, Sector to Rows and SUM(Units_Consumed) to Color and Label; select Square in the Marks card.
6. Add the title “Units Consumed by Sector and Region”.
7. Right-click Month and Sector in the Data pane (or in the view) and choose Show Filter; set both filters to apply to all worksheets using this data source.
8. Create a new Dashboard and add the line chart and heatmap, together with the Month and Sector filters.
9. Add the title “Energy Consumption Dashboard”.
10. Use the heatmap or the line chart as a filter (click “Use as Filter”) to make the dashboard interactive.

11. Test the dashboard by selecting a month, sector or region and checking the linked views.



Output 4: Energy Consumption Dashboard

Observation: The Tableau dashboard combines the monthly energy usage line chart and the Sector vs Region heatmap with Month and Sector filters, allowing interactive exploration of energy consumption.

Procedure

1. Save the dataset as Energy_Consumption_Data.csv and import it into R using read.csv().
2. Plot the histogram with a density curve and the density plot of Units_Consumed, and compare the consumption by sector using aggregate().
3. Create the scatter plot of Temperature and Units_Consumed with bubble size for Peak_Hours and transparency using ggplot2.
4. Calculate the average Renewable_Usage for each Sector using aggregate() and create the bar chart using barplot().
5. Open Tableau and import Energy_Consumption_Data.csv.
6. Create the monthly energy usage line chart and the Sector vs Region heatmap in Tableau.
7. Add the Month and Sector filters.
8. Combine the visualizations into an interactive dashboard and add a suitable title.

9. Test the dashboard using the filters and interactivity.

Result

The Energy Consumption data was successfully imported into R and analysed using a histogram, density plot, bubble scatter plot and renewable usage bar chart, and visualized in an interactive Tableau dashboard.